

Biosynthesis Of Silver Nanoparticles from selected Cyanobacteria and their Antibacterial Potential

Abstract

The emergence of multi drug-resistant (MDR) bacterial infections and cancer has necessitated the development and discovery of alternative eco-safe antibacterial and anticancerous agents. Biogenic fabrication of metallic nanoparticles is an emerging discipline for production of nanoproducts that exert potent antibacterial activity. Silver nanoparticles produced from aqueous cyanobacterial extracts is seen as a green, eco-friendly, low-cost biotechnological novelty, that outperforms chemical and physical approaches. In this current study, aqueous extracts of eight different cyanobacteria, viz. *Anabaena* sp. [Sub4 (-N)], *Mastigocladus* sp. (74-N), *Nostoc* sp. (222-N), *Anabaena* sp. (46-N), *Oscillatoria* sp., *Lyngbya majuscula*, *Lyngbya aestuarii* and *Anabaena* sp. (31-N) were made from fresh biomass and then used for the green synthesis of AgNPs. The UV-Visible absorption spectrum was used to confirm the biosynthesis of AgNPs. In this study, synthesized AgNPs were tested for resazurin based antibacterial test against six multidrug-resistant bacteria— *Bacillus cereus* (ATCC 13061), *Bacillus subtilis* (MTCC 121), *Listeria monocytogenes* (MTCC 657), *Staphylococcus aureus* (MTCC 96), *Salmonella typhi* (MTCC 98) and *Escherichia coli* (MTCC 1667). The AgNPs synthesized from seven organisms out of the eight different cyanobacteria used, were found to have potent antibacterial activities against those pathogenic bacteria. So, AgNPs can be used as antibacterial agents.

1. Introduction

The most sought-after field of study nowadays is Nanotechnology, owing to the ubiquitous potential and application of nanoparticles in numerous physical, biological, biomedical, biotechnological, pharmaceutical and cosmetics fields. A nanoparticle, defined by the European Commission, is a small particle that ranges between 1 to 100 nanometres in size and behaves as a whole unit in terms of its transport and properties.[1] Undetectable to the human eye, nanoparticles exhibit significantly different physical and chemical properties to their larger material counterparts. The material properties change as their size approaches the atomic scale. Owing to their very small size, nanoparticles have a very large surface area to volume ratio when compared to bulk material.[2] This feature enables nanoparticles to possess unexpected optical, electronic, magnetic, physical and chemical properties, as they are small enough to confine their electrons and produce quantum effects.

Nanoparticles have garnered huge interest in medical field for their ability to deliver drugs directly to the site of target in the ideal dosage range, increasing therapeutic efficiency of the drugs, debilitating side effects and improving patient compliance.[2][3] Use of nanoparticles for biological and cell imaging, photo thermal therapy, detection of proteins or pathogens and tissue engineering is also being investigated.[2][3][4] The antineoplastic effect of nanoparticles is also being researched for effective inhibition of tumour growth.[5] Besides this, silver nanoparticles are being used increasingly in wound dressings, catheters and various household products due to their antimicrobial properties.[6] Antimicrobial properties of inorganic nanoparticles have increased the potential application of nanoparticles in textile, water disinfection and food processing and packaging.[7] The unique physico-chemical properties of the nanoparticles make them very desirable material for mass-production of novel marketable goods like microelectronics, miniature chips and nanodevices, as well as in aerospace and food industries.[8][9][10] The inimitable plasmon absorbance features of noble metal nanoparticles have been exploited for a wide variety of applications, including chemical sensors and biosensors.[3][11][12]

Artificial nanoparticles can be created from any solid or liquid material, including metals, dielectrics and semiconductors, as well as biological specimens.[13] There are several methods for creating nanoparticles, including gas condensation, attrition, chemical precipitation, ion implantation, pyrolysis, hydrothermal synthesis and biosynthesis.[1][13] However, exploiting the physical and chemical routes lead to high energy consumption, low yield, high cost and environmental damage due to generation of toxic by-products.[1][2][6] The biological pathway, i.e., synthesis of nanoparticles from microorganisms (bacteria, fungi, yeast, algae, etc.) or plants, is way more lucrative. It is a high yielding, non-toxic, cost-efficient, low maintenance and

environmentally friendly alternative to the physical or chemical methods.[1][7][14] In recent years, an emerging trend of synthesizing nanoparticles from algae is developing, because of the ability of algae to accumulate metals and reduce metal ions.[14] This has opened up new dimensions for study of algae-mediated biosynthesis of economically valuable nanometals and their immense potential and applications. This newly evolving branch of science has been tentatively dubbed as Phyconanotechnology.[7][14]

Till now, different groups of algae such as Cyanophyceae, Chlorophyceae, Phaeophyceae, Rhodophyceae and others (diatoms and euglenoids) have been reported to be used for biosynthesis of metallic nanoparticles.[1][14] Whole cells of *Plectonema boryanum* UTEX 485 (filamentous cyanobacteria) were reported to be efficient in the production of Pt, Ag, and Pd nanoparticles.[14][15] The cyanobacterial strains of *Anabaena flos-aquae* and *Calothrix pulvinata* were used to implement the biosynthesis of Au, Ag, and Pt nanoparticles.[14] Several algae such as *Lyngbya majuscula*, *Spirulina platensis*, *Chlorella vulgaris*, *Oscillatoria willei* NTDM01, *Gloeocapsa* sp. and *Phormidium valderianum*, were reported to be used as a cost effective method for Ag nanoparticle synthesis.[1][6][14][15] *Spirulina platensis* was also reported to be used for the extracellular synthesis of Au, Ag, and Au/Ag bimetallic nanoparticles.[14] Synthesis of extracellular Ag nanoparticles using *Sargassum wightii* and *Kappaphycus alvarezii* has also been reported.[7][14] Bioreduction of Au (III)-Au (0) using the brown alga, *Fucus vesiculosus*, and biosynthesis of Au nanoparticles using red algae *Chondrus crispus* and green algae *Spyrogira insignis* have been reported recently.[14] Synthesis of gold nanoparticles and nanoplates at room temperature using *Shewanella* algal cell extract has also been reported.[7] In addition to seaweeds, microalgae such as diatoms (*Navicula atomus* and *Diadesmis gallica*) have also been confirmed to have the potential to produce Au nanoparticles and silica-gold bionanocomposites.[1]

Silver has long been valued as a precious metal. It exhibits the highest electrical conductivity, thermal conductivity and reflectivity of any metal. It is also highly recognized for its antimicrobial properties against bacteria, fungi and certain viruses. As a result, production of silver nanoparticles (AgNPs) from different algal sources is attracting immense attraction due to their varied properties, like magnetic and optical polarizability, catalytic, therapeutic and antimicrobial activities.[15] Distinctive antibacterial and antifungal properties of AgNPs make suitable for use in medical, pharmaceutical, textile and food industry, as well as water treatment facilities for water purification.[7] It has also been reported to have diversified use in day-to-day household products like food storage containers, cellular phones, socks, slippers, pillows, wet wipes, detergents, soaps, shampoos, tooth pastes, air filters, respirators, coatings of refrigerators, vacuum cleaners, washing machines, air sanitizer sprays, etc.[15] AgNPs biosynthesized from certain macro-algae (*Turbinaria turbinata*) and micro-algae (*Anabaena oryzae*, *Nostoc*

muscorum and *Calothrix marchica*), show promising anti-cancerous activities.[6] AgNPs have also been reported to be used for mass spectrometry of peptides and colorimetric determination of histidine and ammonia.[15]

This study focuses on the biosynthesis of AgNPs from 8 different cyanobacterial strains following a partially modified method adopted by Lengke et al. 2007 and their characterization, as well as study of their antibacterial activities.[15]

2. Materials and Methods

2.1. Cyanobacteria Isolation and Culture

Eight cyanobacterial species, isolated from various environments, were taken from the Visva Bharati Culture Collection of Algae (VBCCA) for biosynthesis of AgNPs.

Borosil glass test tubes (15x150mm), petridishes and conical flasks (250ml and 100ml) were used for culturing the isolated cyanobacteria. The glasswares were firstly cleaned using liquid detergent (Teepol) in running tap water and then dried. The dried glasswares were immersed in Chromic Acid (Potassium di-chromate in concentrated Sulphuric Acid, 1% W/V), followed by repeated washing in tap water and then finally washed in double distilled water.

BG-11 culture medium, with or without nitrogen, was used for culturing the organisms.[16] The following chemicals (BDH, Merk, HiMedia and SRL) were added at specific concentration to 1 litre of double distilled water for preparation of the culture media [16]:-

Chemical Constituents	Concentration (g/l)	Stock (g/per 100ml)
1. MgSO ₄ .7H ₂ O	0.075	7.5
2. CaCl ₂	0.036	3.6
3. Na ₂ CO ₃	0.02	2
4. Micronutrients		
i. H ₃ BO ₃	2.86	286
ii. MnCl ₂ .4H ₂ O	1.81	181
iii. ZnSO ₄ .7H ₂ O	0.222	22.2
iv. CuSO ₄ .5H ₂ O	0.079	7.9
v. Na ₂ MoO ₄ .2H ₂ O	0.039	3.9
vi. CO(NO ₃) ₂ .6H ₂ O	0.049	4.9
5. i. Citric Acid	0.006	0.6
ii. Ferric Ammonium Citrate	0.006	0.6
iii. EDTA	0.001	0.1
6. K ₂ HPO ₄	0.04	4
7. NaNO ₃ (for N+ medium)	1.5	150

Approximately 800ml of double distilled water was taken in a 1 litre graduated round bottom flask. 1ml of each stock solution of the chemical constituents was added sequentially into the flask with a time interval of 5 minutes with repeated stirring. Only K₂HPO₄ was added after 30 minutes of adding the EDTA solution. After addition of all the ingredients, the volume of the solution was made up to 1L by adding double distilled water.

The pH of the medium was maintained at 7.4- 7.8 before autoclaving.

Before sterilization, the test tubes and flasks were plugged with non-absorbent cotton. The culture medium and the glasswares were sterilized at 20lb pressure for 30 minutes in the autoclave and then immediately put into the laminar air flow where they were irradiated with UV-C (main output at 254nm) for 24 hours.

Inoculation was done in the laminar air flow while maintaining aseptic conditions throughout. A homogenous suspension of each organism was made using a sterile glass tissue homogenizer of 20ml capacity. Equal amount of each inoculum was pipetted into each culture vessel, i.e., conical flasks containing the sterile culture medium, by using a micropipette.

The culture vessels were maintained in the culture racks in a temperature controlled room at 25±1°C temperature under light intensity of 7.5w/M² emitted from daylight fluorescent tubes over a 12 hours light/dark cycle. The culture vessels were hand-shaken twice daily to prevent sticking of the algal cells to the glass wall of the culture vessels. The volume of the liquid culture medium in the culture vessels was always maintained at half of the culture vessel to overcome contamination due to splash during shaking.

2.2. Maintenance of Culture

The stock cultures of the isolated organisms were maintained in agar slants in glass test tubes containing 1.1% (W/V) agar in the basal inorganic medium. Each organism was transferred from the agar slants into 100ml capacity conical flask containing 50ml fresh sterile medium. The organisms were grown under the above mentioned conditions to ensure rapid growth. These cultures were used for the experiment once they had reached exponential growth phase.

2.3. Identification of the Cyanobacterial Strains

The cultured organisms were observed under Leica DM 750 trinocular microscope and micro-photographs of each of the cyanobacterial strains were taken using Leica EC3 scientific digital camera and analyzed with LAS EZ software. The organisms were then identified using “Cyanophyta- I.C.A.R. Monographs on Algae (1959)” by Dr. T.V. Desikachary.

The cyanobacterial strains (**Fig. 1**) used in this study are as follows:-

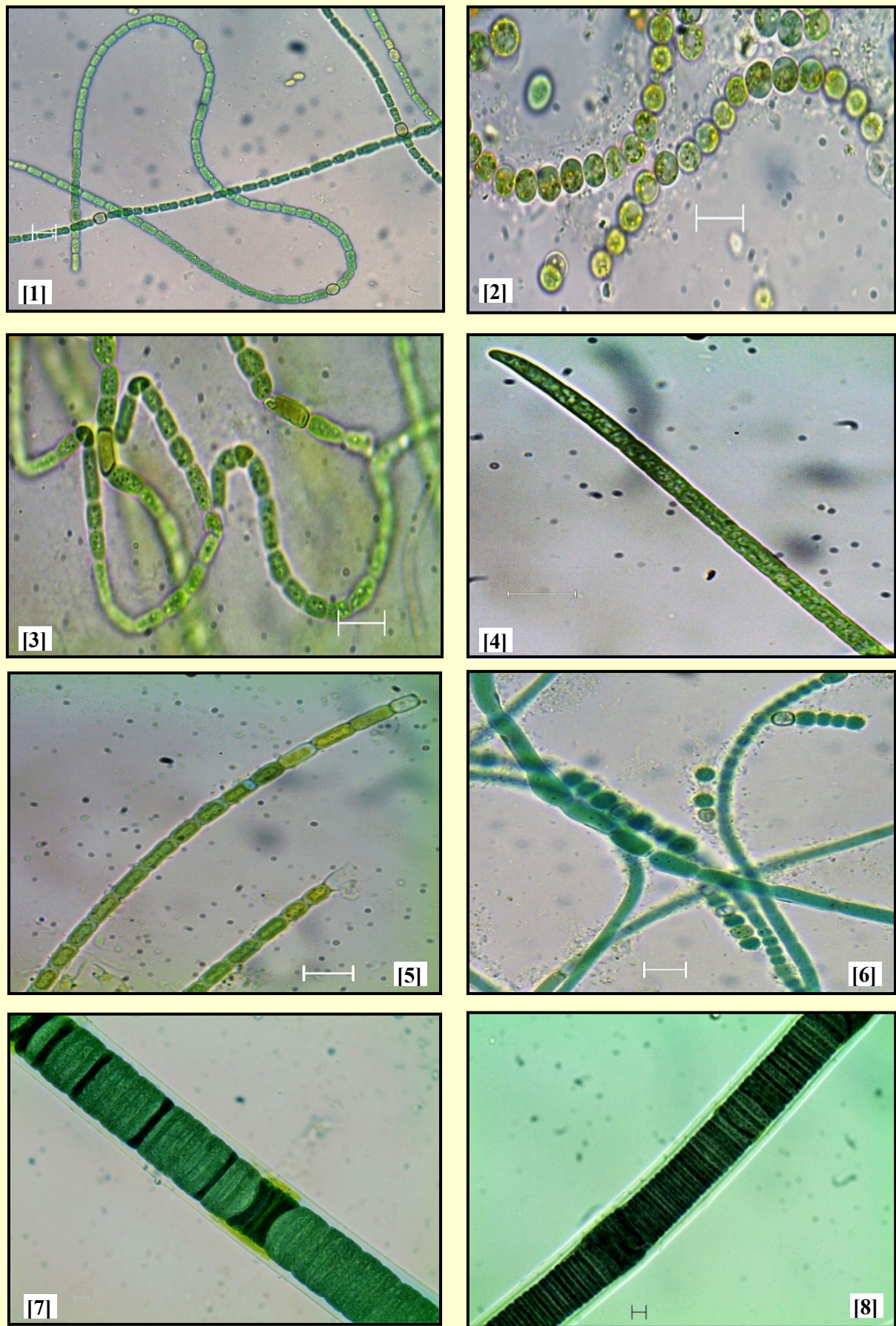


Fig. 1: The cyanobacterial strains observed under light microscope— [1] *Anabaena* sp. 31-N, [2] *Nostoc* sp. 222-N, [3] *Anabaena* sp. 46-N, [4] *Oscillatoria* sp., [5] *Anabaena* sp. SUB-4 (-N), [6] *Mastigocladus* sp. 74-N, [7] *Lyngbya aestuarii* and [8] *Lyngbya majuscula*

2.4. Biosynthesis of AgNPs

The cultured biomasses were centrifuged at 10000 rpm for 12 minutes and the pellets were air dried. The dried specimens were crushed to form fine powder. 10ml distilled water was added to 10mg of each powdered specimen and kept in an incubator at 40°C for 24 hours (Fig. 2).



Fig. 2: Water extract of the Cyanobacterial Strains— *Anabaena* sp. 46-N, *Nostoc* sp. 222-N, *Anabaena* sp. SUB-4 (-N), *Oscillatoria* sp., *Mastigocladus* sp. 74-N, *Anabaena* sp. 31-N, *L. aestuarii* and *L. majuscula* (from left to right) —after 24 hours of incubation at 40°C and before addition of AgNO₃

After 24 hours, the specimens were taken out and filtered using Whatman filter paper. 90ml of AgNO₃ (1mg/2ml) solution was mixed with each filtrate and maintained at 30°C for 48 hours in light. A control was also prepared by mixing 10ml distilled water with 90ml of AgNO₃ (1mg/2ml) solution and maintained at 30°C for 48 hours in light. Synthesis of AgNPs were noticed when the solutions gradually started to turn brown (Fig. 3).



Fig. 3: *Mastigocladus* sp. 74-N, *Anabaena* sp. SUB-4 (-N), *L. majuscula*, *Anabaena* sp. 31-N, *Anabaena* sp. 46-N, *Oscillatoria* sp., *Nostoc* sp. 222-N and *L. aestuarii* (from left to right) after 48 hours of adding AgNO₃ solution and maintaining at 30°C

Spectrum of the solutions were taken in UV-Vis spectrophotometer (UV-1800 240V, Shimadzu Corporation, Kyoto, Japan) to ascertain the production of AgNPs from the various cyanobacterial strains.

After taking spectrum, the solutions were centrifuged at 10000 rpm for 15 minutes. The pellets obtained were carefully scraped to obtain fine powdered AgNPs, which were stored for further analysis (Fig. 4).



Fig. 4: Powdered AgNPs produced from the different cyanobacterial strains

2.5. Antibacterial assay using Microtiter plate

A resazurin microtiter plate-based assay was conducted to ascertain antibacterial properties of the biosynthesized AgNPs, following a modified method [17]. A microtiter plate containing 96 wells was used under aseptic conditions for the assay. All the purified and dried AgNPs biosynthesized from the eight different cyanobacteria were screened for antibacterial test against six pathogenic bacterial cultures— *Bacillus cereus* (ATCC 13061), *Bacillus subtilis* (MTCC 121), *Listeria monocytogenes* (MTCC 657), *Staphylococcus aureus* (MTCC 96), *Salmonella typhi* (MTCC 98) and *Escherichia coli* (MTCC 1667), which were procured from Microbial Type Culture Collection (MTCC), IMTech, Chandigarh.

The powdered Ag-NPs were firstly dissolved in DMSO (does not have any inhibition property to prevent bacterial growth) to make a testing concentration of 4 mg/ml. Bacteria were grown overnight and diluted later by sterile nutrient broth and a specific OD of 0.00075 was measured in a spectrophotometer. 97.5µL of bacterial suspension was added in each well. 2.5µL of each of the test material solution was added to the respective wells, making the final test volume in each well 100µL.

Antibiotic Ciprofloxacin was used as a positive control. To check sterility control, sterile nutrient broth was added instead of bacterial suspension to check whether the test materials were sterile. For negative control, only bacterial suspensions were added to the wells to check whether the bacteria were growing or not. To check whether the solvent (DMSO) was inhibiting growth of bacteria or not, another control was used with DMSO and bacterial suspension.

Resazurin solution was prepared at 0.01% concentration with double distilled water. A volume of 4µL of Resazurin was added to each well and mixed thoroughly. Upper lid of the plate was covered after the completion of all the process for maintaining contamination free environment. Finally the plate was incubated at 37°C under frequent observation to detect any visual color change.

3. Results and Discussions

Formation of AgNPs was indicated by color change from pale yellow to dark brown.[19] The change in color occurred because of the vibrations of surface plasmons in the produced AgNPs.[20] Surface Plasmon Resonance peak is an indicator of the size and monodispersity of the nanoparticles, and absorbance peak in the range of 320–580 nm is usually typical for AgNPs.[6][15][18] Previous studies have shown that the absorption peak of AgNPs synthesized from *Oscillatoria willei* was found at 450 nm and that produced from *Oscillatoria limnetica* was found at 426 nm.[6][18] Another study showed that absorbance peaks of AgNPs synthesized from *Lyngbya* sp. (NCCU-102), *Oscillatoria* sp. (NCCU-369), *Anabaena ambigua* (NCCU-1), *Anabaena variabilis* (NCCU-441) and *Nostoc muscorum* (NCCU-442) were found at 452 nm, 485 nm, 446 nm, 450 nm and 443 nm respectively. [15]

The peak absorbances observed in the spectrophotometer are recorded in a tabular form (**Table 1**) below, along with the obtained graphs (**Fig. 5- Fig. 12**):-

Organisms	Wavelength (nm)	Absorbance (Peak)
<i>Anabaena</i> sp. 31-N	432	0.884
<i>Nostoc</i> sp. 222-N	429	0.493
<i>Anabaena</i> sp. 46-N	—	—
<i>Oscillatoria</i> sp.	—	—
<i>Anabaena</i> sp. SUB-4 (-N)	429.50	0.297
<i>Mastigocladus</i> sp. 74-N	447	0.630
<i>Lyngbya aestuarii</i>	460	1.322
<i>Lyngbya majuscula</i>	434	0.604

Table 1: Showing wavelength of peak absorbance for the different biosynthesized AgNPs

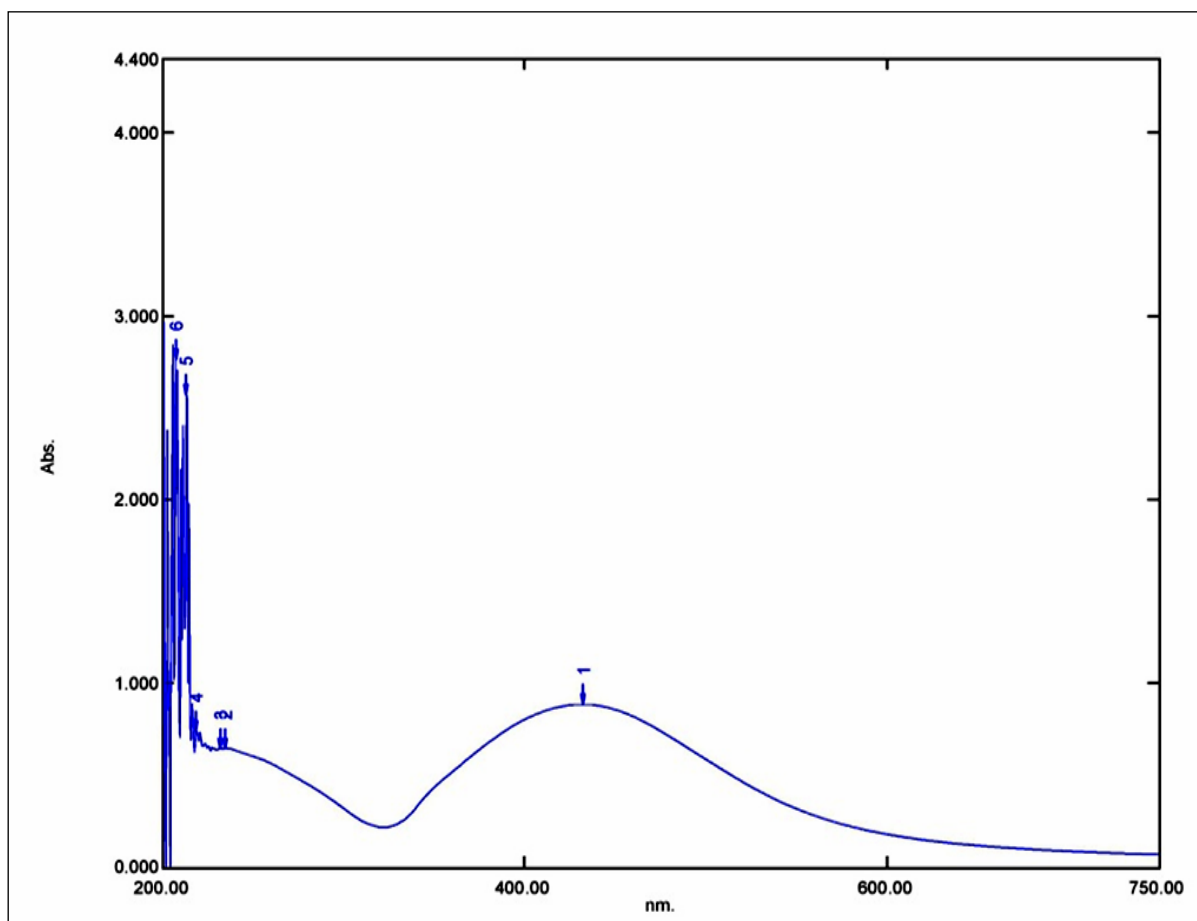


Fig. 5: UV-visible absorption spectrum of AgNPs biosynthesized from *Anabaena* sp. 31-N

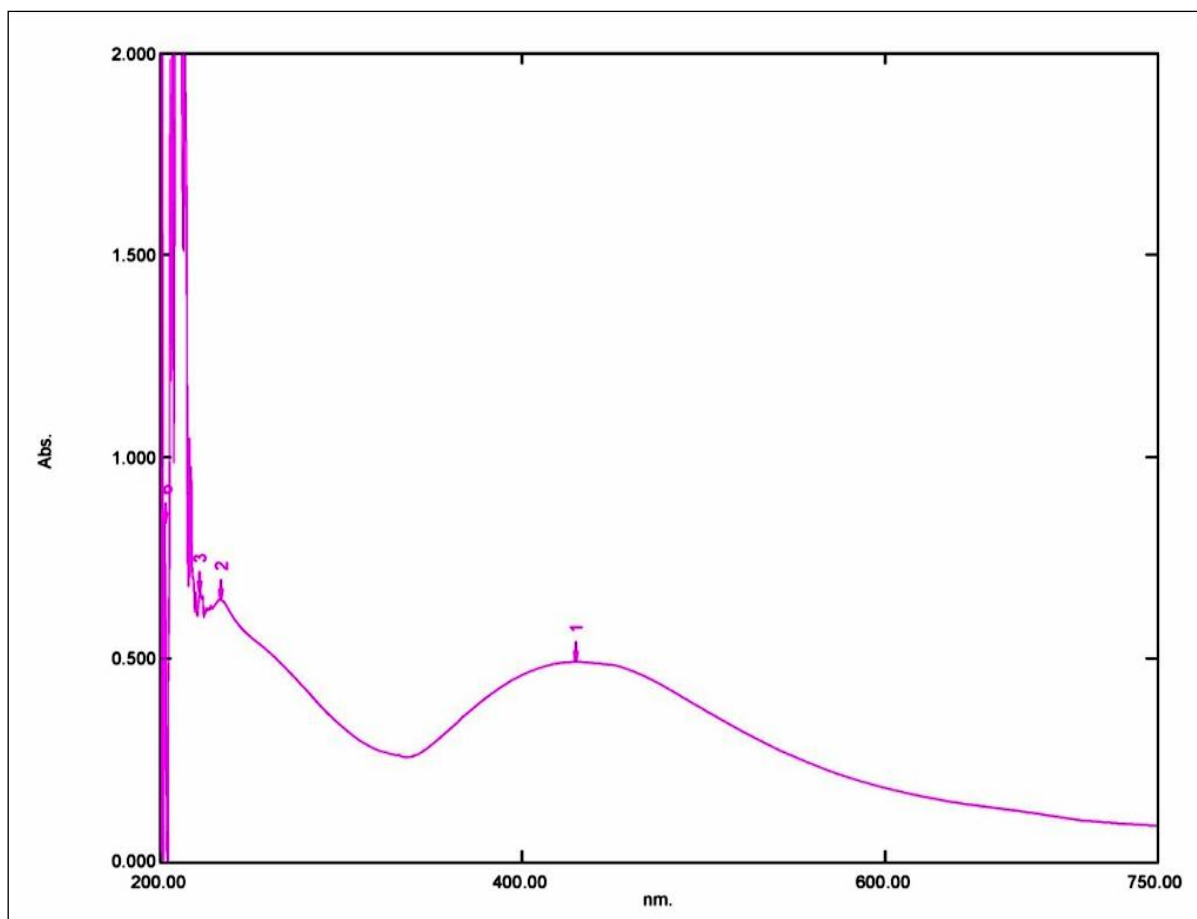


Fig. 6: UV-visible absorption spectrum of AgNPs biosynthesized from *Nostoc* sp. 222-N

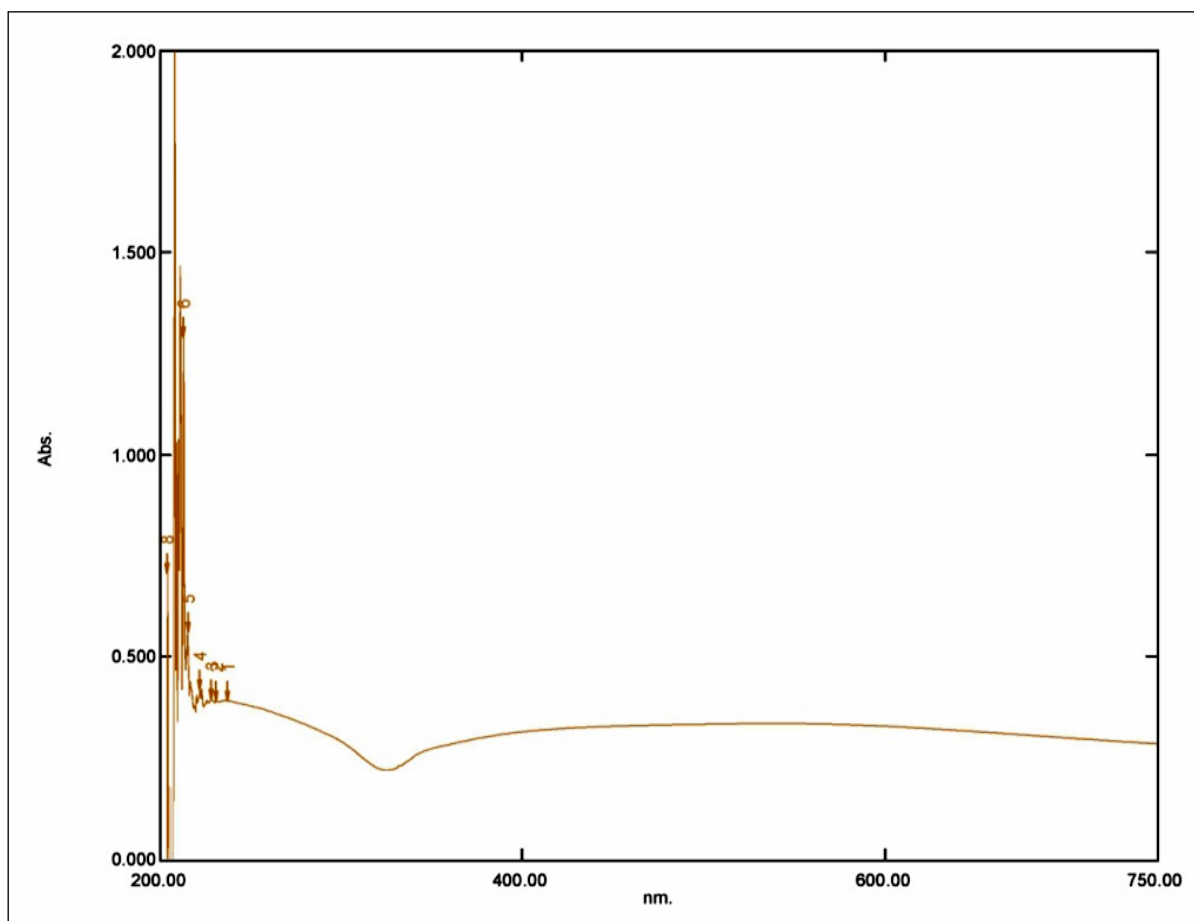


Fig. 7: UV-visible absorption spectrum of AgNPs biosynthesized from *Anabaena* sp. 46-N

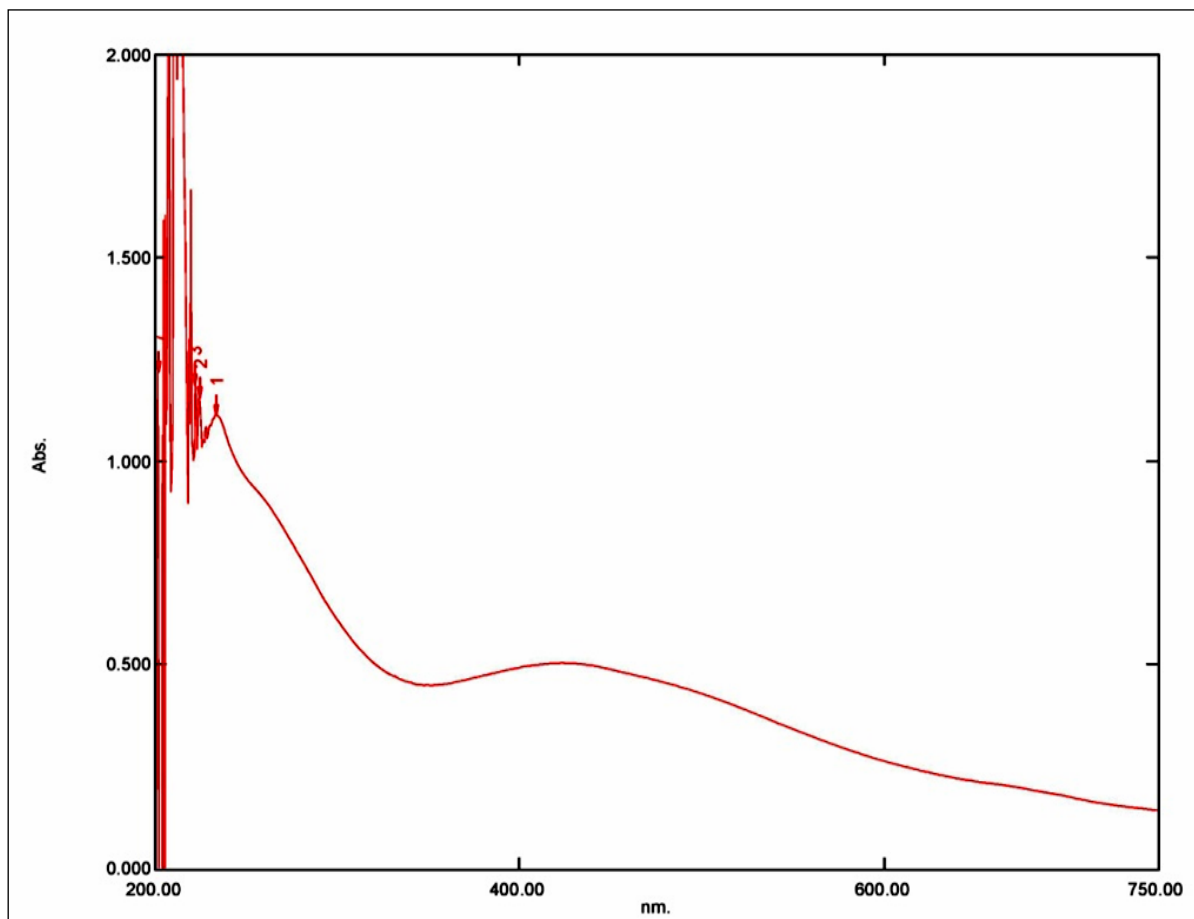


Fig. 8: UV-visible absorption spectrum of AgNPs biosynthesized from *Oscillatoria* sp.

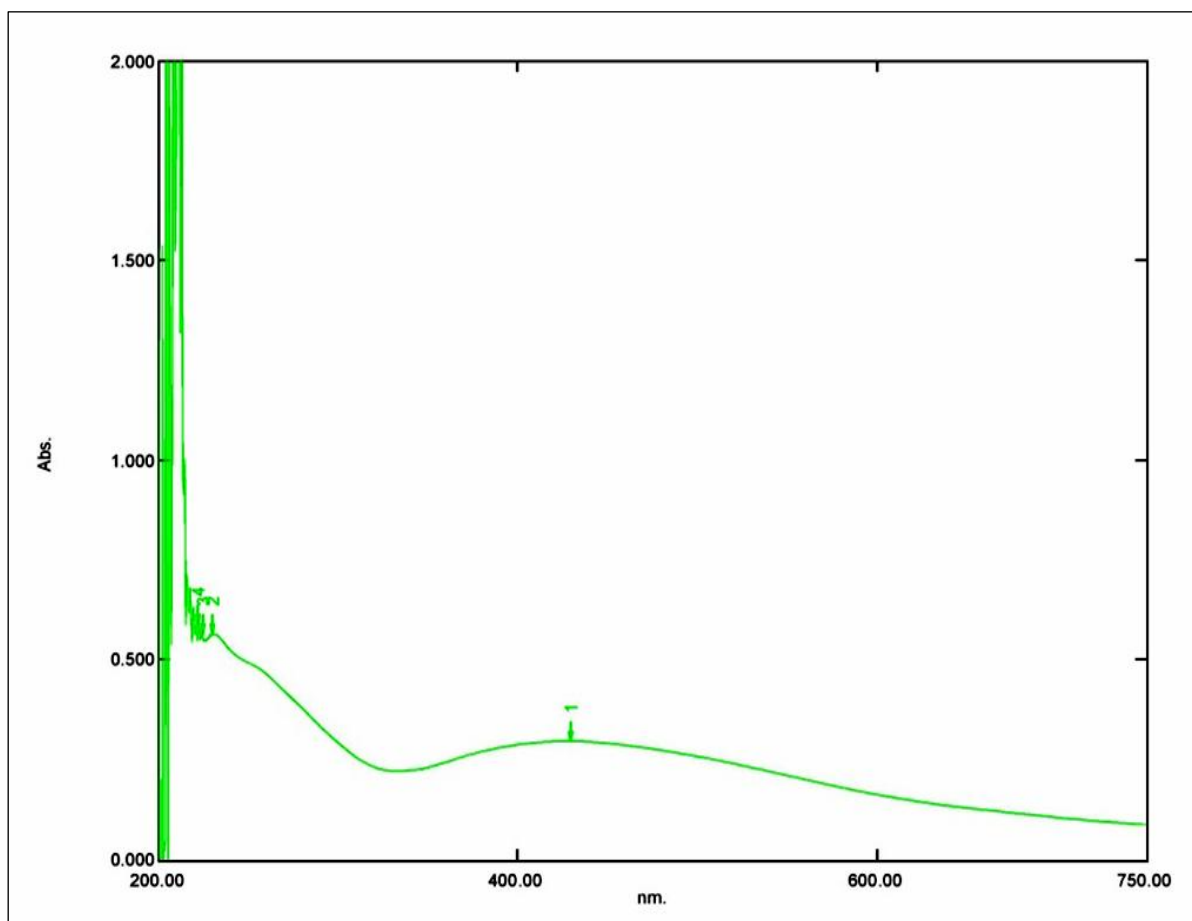


Fig. 9: UV-visible absorption spectrum of AgNPs biosynthesized from *Anabaena* sp. SUB-4 (-N)

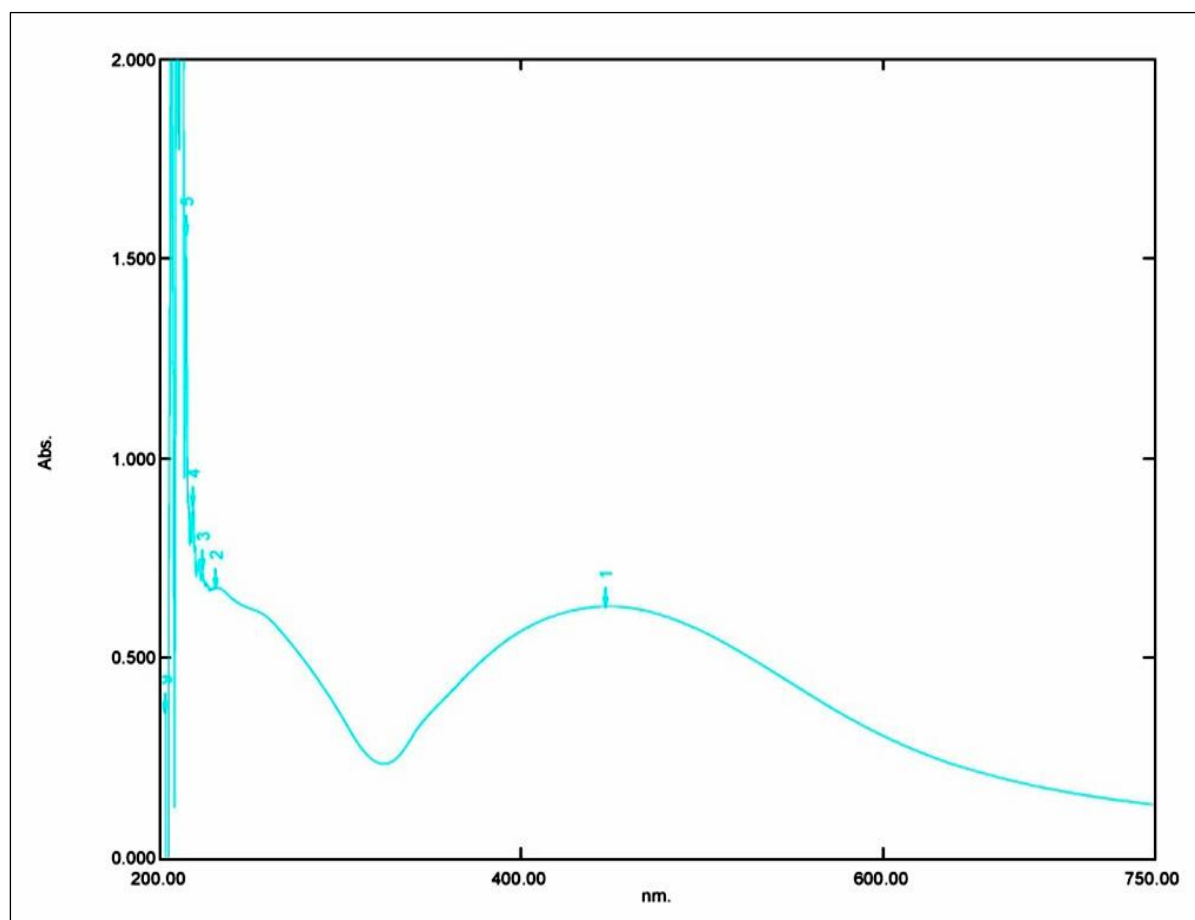


Fig. 10: UV-visible absorption spectrum of AgNPs biosynthesized from *Mastigocladus* sp. 74-N

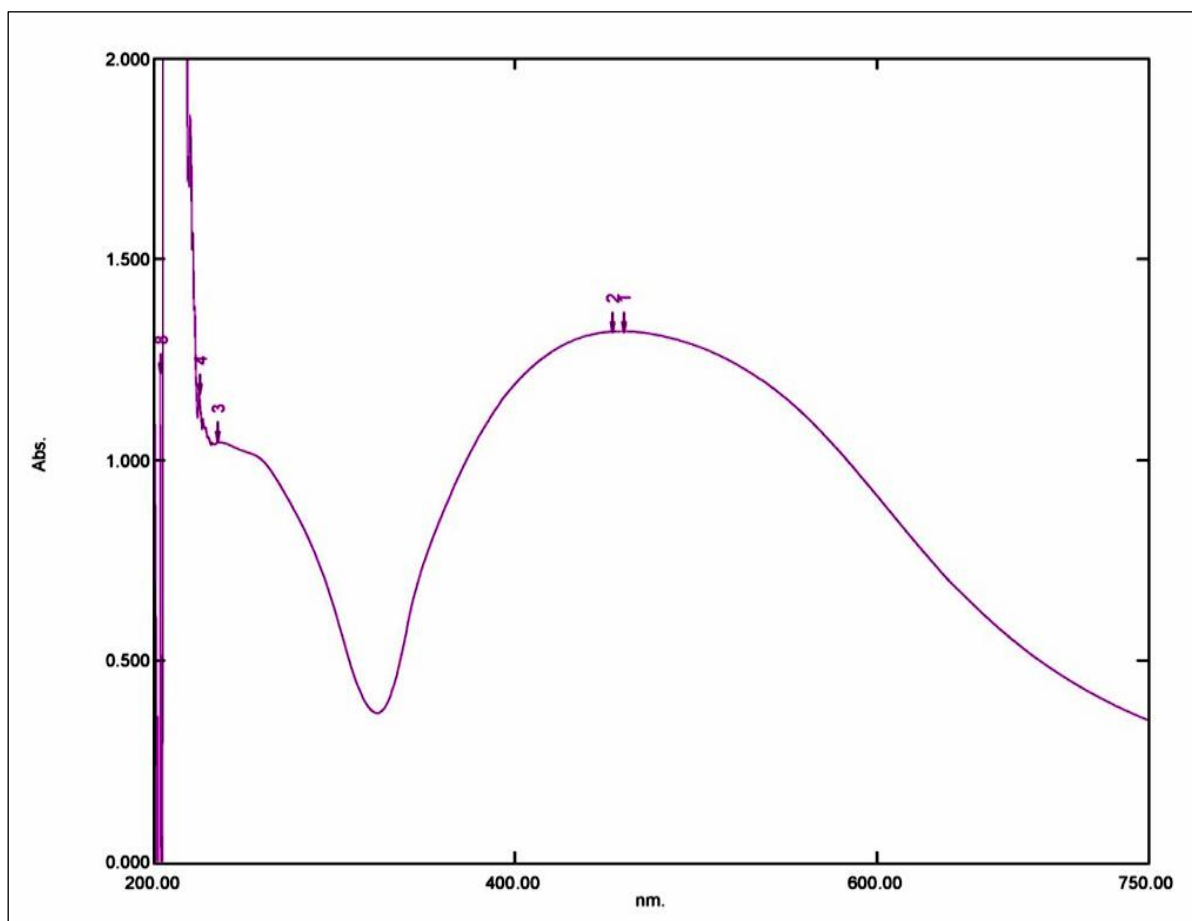


Fig. 11: UV-visible absorption spectrum of AgNPs biosynthesized from *Lyngbya aestuarii*

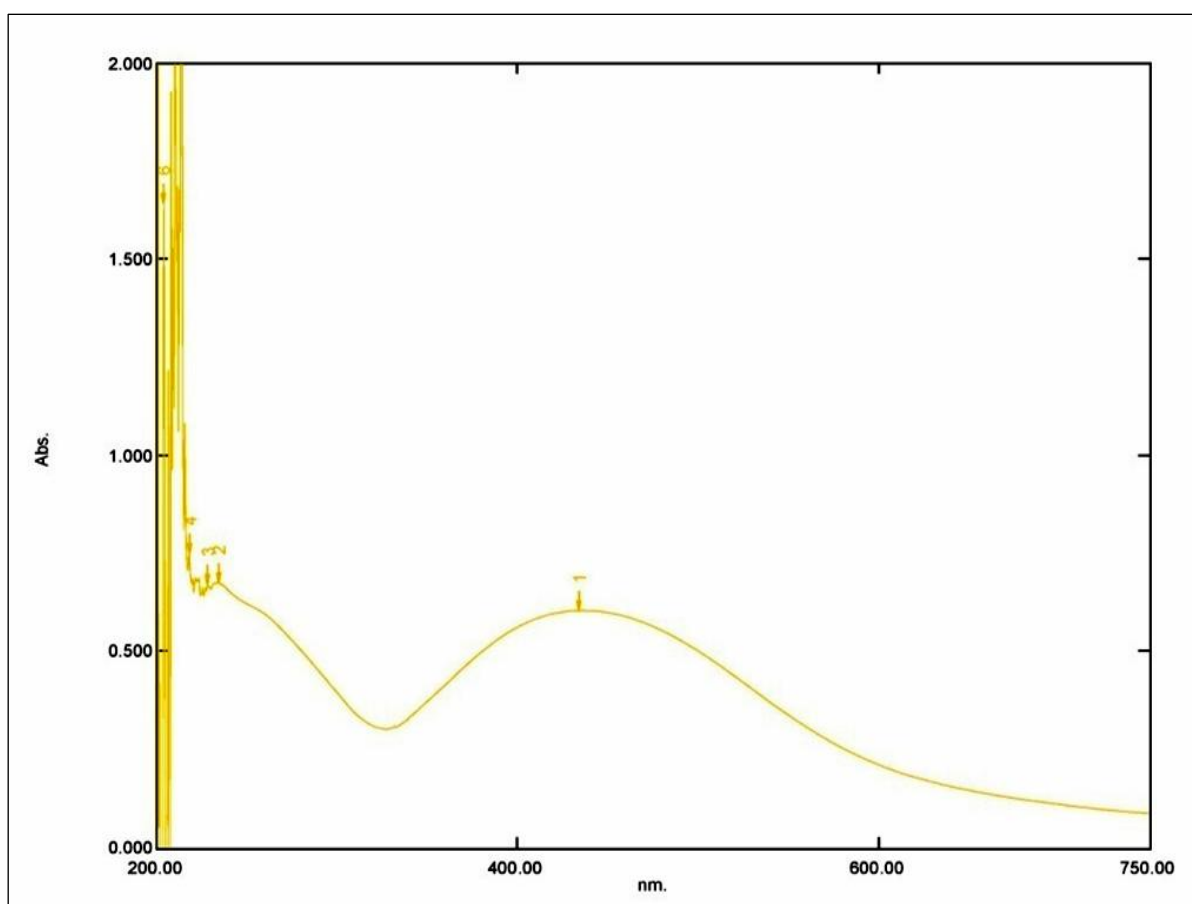
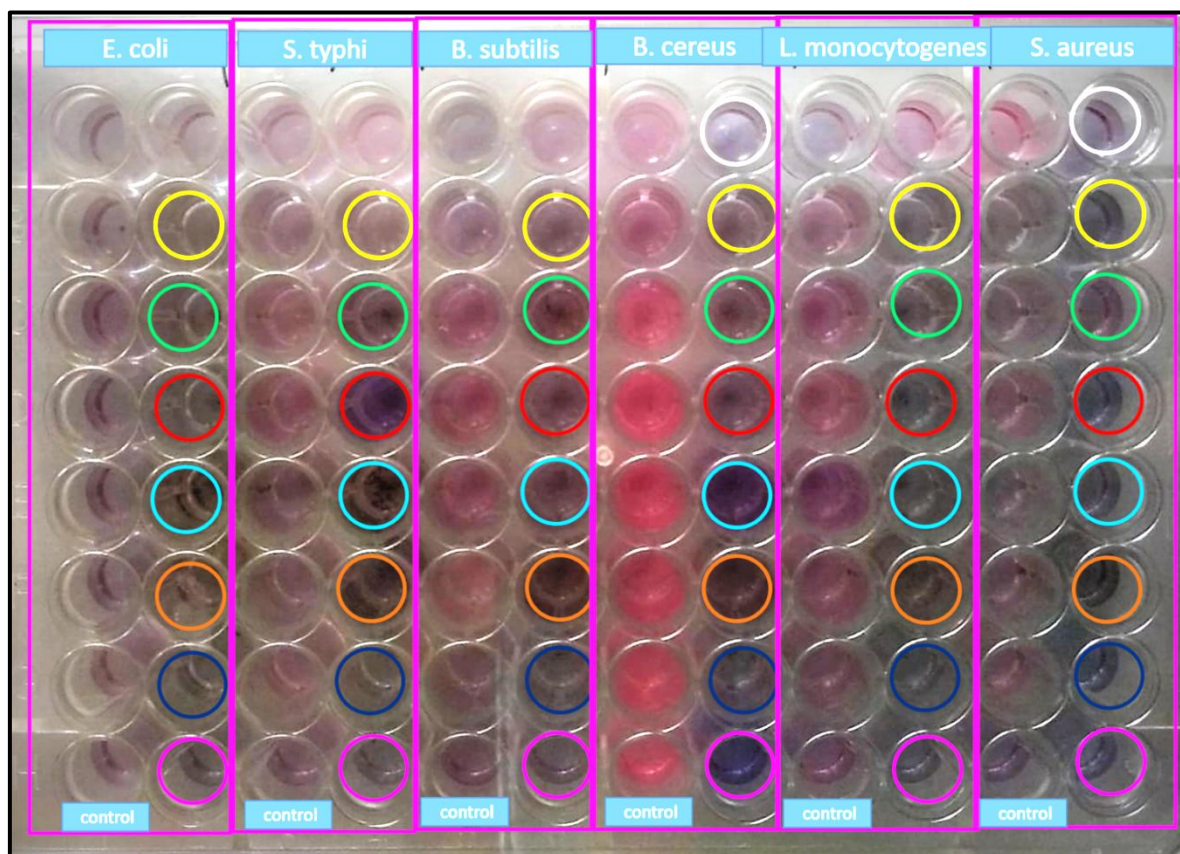


Fig. 12: UV-visible absorption spectrum of AgNPs biosynthesized from *Lyngbya majuscula*

In this study, peak absorbances of the AgNPs synthesized from the eight different cyanobacterial strains were found across the range of 429 nm to 460 nm. This result indicates that AgNPs were successfully synthesized by reducing AgNO₃ and thus it can be inferred that these cyanobacterial strain may be potential candidates for biofabrication of AgNPs.

Biosynthesized AgNPs have garnered a lot of attention in the medical field as potent antibacterial agents. Recent studies have shown that AgNPs biofabricated from *Synechococcus* sp. show antibacterial properties against the bacterial strains *Bacillus subtilis*, *Escherichia coli* and *Staphylococcus aureus*. [21] One study showed that AgNPs synthesized from *Desertifilum* sp. show antibacterial properties against five bacterial strains— *Shigella flexneri*, *Pseudomonas aeruginosa*, *Bacillus subtilis*, *Bacillus cereus*, and *Salmonella enterica* (clinical isolates), while another study showed that the same show antibacterial activities against methicillin-resistant *Streptococcus aureus* (MRSA), drug-resistant *Klebsiella pneumoniae*, *Escherichia coli*, *Streptococcus mutans*, and *Salmonella typhi* (clinical isolates).[17][22] AgNPs synthesized from *Nostoc linckia* have been found to show antibacterial activities against *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Streptococcus aureus* and *Escherichia coli*. [23] It has also been reported that AgNPs synthesized from *Anabaena* sp. showed strong antibacterial activity against three multidrug-resistant bacteria— *Streptococcus aureus*, *Escherichia coli* and *Klebsiella pneumonia*. [24] Antibacterial properties have also been observed in AgNPs synthesized from *Oscillatoria* sp. against bacterial strains *Escherichia coli*, *Klebsiella* sp., *Salmonella* sp., and *Pseudomonas* sp. [25]

Microtiter plate assay was performed in this study to determine antibacterial activity of the biosynthesized AgNPs. Antibacterial activity of the AgNPs was confirmed when the solutions in the wells containing AgNPs did not change color from blue to pink. Change of color from blue to pink occurs due to reduction of the blue-colored Resazurin by the bacterial enzymes, which indicate bacterial growth. Real-time image of the microtiter plate (**Fig. 13**) is presented below indicating the antibacterial activities of the AgNPs against the bacterial test strains:-



Index	
○ AgNPs synthesized from <i>Anabaena</i> sp. (31-N)	○ AgNPs synthesized from <i>Lyngbya majuscula</i>
○ AgNPs synthesized from <i>Anabaena</i> sp. [SUB4 (-N)]	○ AgNPs synthesized from <i>Lyngbya aestuarii</i>
○ AgNPs synthesized from <i>Mastigocladus</i> sp. (74-N)	○ AgNPs synthesized from <i>Nostoc</i> sp. (222-N)
○ AgNPs synthesized from <i>Anabaena</i> sp. (46-N)	○ AgNPs synthesized from <i>Oscillatoria</i> sp.

Fig. 13: Microtiter plate showing antibacterial activity of the synthesized AgNPs against the test bacterial strains (Index shows the color of circle the particular AgNP represents)

In this study, antibacterial activity was observed in the AgNPs biosynthesized from the cyanobacterial strains— *Anabaena* sp. [Sub4 (-N)], *Mastigocladus* sp. (74-N), *Nostoc* sp. (222-N), *Anabaena* sp. (46-N), *Oscillatoria* sp., *Lyngbya majuscula*, *Lyngbya aestuarii* and *Anabaena* sp. (31-N), through microtiter plate assay. From the plate assay, it was concluded that all the biogenic AgNPs showed antibacterial properties against all the six test bacterial strains— *Bacillus cereus* (ATCC 13061), *Bacillus subtilis* (MTCC 121), *Listeria monocytogenes* (MTCC 657), *Staphylococcus aureus* (MTCC 96), *Salmonella typhi* (MTCC 98) and *Escsherichia coli* (MTCC 1667), except AgNPs synthesized from *Anabaena* sp. (31-N) which only showed antibacterial activity against *Bacillus cereus* (ATCC 13061) and *Staphylococcus aureus* (MTCC 96) (Table 2).

Organisms from which AgNPs were biosynthesized	Working concentrations of the AgNPs	<i>E. coli</i> Gram (-)ve	<i>S. typhi</i> Gram (-)ve	<i>B. subtilis</i> Gram (+)ve	<i>B. cereus</i> Gram (+)ve	<i>L. monocytogenes</i> Gram (+)ve	<i>S. aureus</i> Gram (+)ve
<i>Anabaena</i> sp. (31-N)	100 µg/ml	No	No	No	Yes	No	Yes
<i>Anabaena</i> sp. [SUB4 (-N)]	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Mastigocladus</i> sp. (74-N)	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Anabaena</i> sp. (46-N)	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Lyngbya majuscula</i>	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Lyngbya aestuarii</i>	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Nostoc</i> sp. (222-N)	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes
<i>Oscillatoria</i> sp.	100 µg/ml	Yes	Yes	Yes	Yes	Yes	Yes

Table 2: Showing antibacterial activity of the AgNPs against the test bacterial strains

Thus, it can be inferred that the AgNPs synthesized from these cyanobacterial strains can be considered as promising antibacterial agents against certain pathogenic bacteria. Their potent inhibitory effects against bacterial growth make them highly compatible to be used as an alternative to the more well-known antibiotics.

4. Conclusion

Cyanobacteria, the oldest and most diverse phylum of photoautotrophic prokaryotes, are the first known organisms to have produced oxygen and are responsible for the transition of the primordial carbon dioxide-rich atmosphere of the Earth to the relatively oxygen-rich contemporary atmosphere. In recent days, Cyanobacteria have been found to be the source of novel biologically active secondary metabolites which have antibacterial, antiviral, antifungal, antioxidative and anticancer properties.

In this study, we biosynthesized AgNPs from eight different cyanobacterial strains— *Anabaena* sp. [Sub4 (-N)], *Mastigocladus* sp. (74-N), *Nostoc* sp. (222-N), *Anabaena* sp. (46-N), *Oscillatoria* sp., *Lyngbya majuscula*, *Lyngbya aestuarii* and *Anabaena* sp. (31-N), and tested their antibacterial activities against six multi-drug resistant bacterial strains— *Bacillus cereus* (ATCC 13061), *Bacillus subtilis* (MTCC 121), *Listeria monocytogenes* (MTCC 657), *Staphylococcus aureus* (MTCC 96), *Salmonella typhi* (MTCC 98) and *Escherichia coli* (MTCC 1667). We isolated, cultured and identified eight cyanobacteria for our experiment and used their aqueous extract to biosynthesize AgNPs with the help of AgNO₃ solution. Synthesis of the AgNPs was confirmed by UV-Vis spectrophotography which gave SPR peak at a range of 429 nm to 460 nm. The synthesized AgNPs were tested against the bacterial strains by using

microtiter plate assay and were found to have potent antibacterial activities against the test bacteria.

From this study it can be concluded that biosynthesis of silver nanoparticles from cyanobacteria is a cost-effective, eco-friendly and time-saving alternative to the harsher, high maintenance and environmentally damaging physical and chemical methods. For characterization of AgNPs, FTIR, SEM and XRD can be done in the future. Moreover, the antibacterial properties exhibited by the silver nanoparticles synthesized from these cyanobacteria could be used as an alternative antibacterial agent in the future.

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